1st Assessment: Spefication and Design Proposal (15%)

For your first assessment you should submit a specification and design proposal. The purpose of the proposal is to show that you have a clear idea of what the project involves and a well-defined plan for realising it. The document should be concise but clear, following the structure outlined below. Aim for about 5 pages of text and diagrams (not including the title page) and about 1500-2000 word counts. These numbers though are not a fixed limit. There are no penalties for going over or under the count.

Writing the proposal helps both you and your supervisor to understand what the project is all about, and what the end product will be. At the simplest level it needs to answer three broad questions.

- What will you do?
- How will you do it?
- What will it look like?

If your project involves designing and implementing a piece of software (as most projects do), then the proposal should clearly answer these questions. A proposal with more of a research focus will still cover these things but you would describe your experimental process instead. The key thing is that the document should provide **sufficient detail** using **non-specialist language**, so anyone with a general computer science background would know what your project involves.

How to submit?

You should submit your proposal in Canvas in the Assignment section for CA1: **CA1 - Speficification and Design (15%) (https://canvas.liverpool.ac.uk/courses/86905/assignments/318854)**.

Note that only pdf file will be accepted.

Be aware of the **marking guidelines (https://canvas.liverpool.ac.uk/courses/86905/pages/marking-guidelines)** for this assessment. Feedback will be available via the **e-Project System (https://sam.csc.liv.ac.uk/COMP702)** when the work has been marked.

Check the timetable for the deadline.

The structure of your proposal

Your proposal should have the following structure.

Title page

On the first page you should clearly state the title of the project, your name, and your student ID. You don't need to include the names of the markers or any other details.

Statement of ethical compliance

At the start of your proposal you should state the data and participant categories for your project (for example, A0, C2, etc.) with a sentence to confirm that you will follow our **ethical guidance (https://canvas.liverpool.ac.uk/courses/86905/pages/ethical-guidance)** . This can be included on the title page if you like. It's okay if you later change your plans (such as adding an evaluation activity that you didn't mention in this document), provided you update the statement in future assessments and check with your supervisor before starting the activity.

Project description

This should be a high-level overview of your project, suitable for a reader who isn't an expert in the topic area. This will be most useful for the second marker, who might have no prior knowledge of the project. This first section should be relatively short and succinct, but make sure you answer the key question of what you will be doing on the project. You can add more detail in the later sections.

Aims and requirements

The aims of your project are broad statements of intent. In other words, what you expect to have achieved after completing the work. You will probably have two or three aims that capture the overall substance of the project, but you can have more if you like. They should be presented as bullet points so they are easy to read and quickly summarise what you intend to do.

You should also include a list of requirements. In other words, things that your software will do. They can be listed in order of priority, or categorised into essential and desirable items. In most cases you will think up this list on your own or in discussion with your supervisor, but in some cases you might want to conduct a requirements analysis activity with other people to find out what they want your software to do. If you involved other people, just list the requirements in this section and talk about how you conducted the analysis later in the document when you cover project ethics.

Key literature and background reading

During the first few weeks of the project you will be conducting background reading and looking at various sources of information. Some will be technical (such as API documentation) and some will cover previous research in the area. In this section you should demonstrate a good knowledge of the topic area, discussing, citing and summarising key literature, and using it to justify your design and development decisions.

Development and implementation summary

Bearing in mind that you won't have fully started development when you write the proposal, this section should summarise your plans. Provide a brief overview of your proposed development environment and implementation language, and the reasons why you chose them. Each project is different, so you might have little or lots to explain here. Also make sure this section covers the key question of how you will implement (realise) your project, including how your workflow will be organised.

Data sources

If you are using open source datasets, accessing or processing publicly available text, retrieving data from an API, or generating your own data as part of the project, mention it in this section. You should describe the data you will be using, state how it will be obtained, confirm you are using it with permission, and explain how you will ensure confidentiality and anonymity of any personal information. If you're not using or generating any data, you still need to include this section. Check with your supervisor that there is genuinely no data use, and then include a sentence to confirm that this section doesn't apply to your project. Remember that requirements gathering and evaluation activities will generate data.

Testing and evaluation

Explain how you will show that your project works technically (testing) and meets your original requirements (evaluation). At this stage you don't have to include a detailed testing plan, since you probably don't know exactly what components will be produced or how they will work together. However, you should at least have a high-level idea of the type of testing and evaluation you will carry out. If you plan to use other people as beta testers, mention it briefly in this section and then explain what they will do in more detail in the next section.

Project ethics and human participants

In this section you should cover any activities that involve human participants. If you conducted a requirements analysis with other people, describe what you did. If you plan to conduct an evaluation with other people, give a detailed overview of the things you will ask participants to do. Explain how you will anonymise, aggregate and protect participant data. Make sure your plans fall within the permitted activities outlined in our **ethical guidance (https://canvas.liverpool.ac.uk/courses/86905/pages/ethical-guidance)**.

There could be other ethical considerations that are specific to your project, particularly if your supervisor has obtained separate approval for bespoke activity. If so, discuss them with your supervisor and clearly explain them here. If you think your project has no ethical issues and uses no human data or human participants, you still need to include this section. Check with your supervisor that there is genuinely nothing to say, and then include a sentence to confirm that this section does not apply to your project.

BCS project criteria

Your degree programme is accredited by **BCS (https://www.bcs.org/)**, the Chartered Institute for IT. They require honours year projects to demonstrate the following six outcomes.

- *A systematic understanding of knowledge, and a critical awareness of current problems and/or new insights, much of which is at, or informed by, the forefront of the specialist academic discipline*

- *A comprehensive understanding of techniques applicable to their own research or advanced scholarship*
- *Originality in the application of knowledge, together with a practical understanding of how established techniques of research and enquiry are used to create and interpret knowledge in the discipline*
- *Deal with complex issues both systematically and creatively, make sound judgements in the absence of complete data, and communicate their conclusions clearly to specialist and non-specialist audiences*
- *Demonstrate self-direction and originality in tackling and solving problems, and act autonomously in planning and implementing tasks at a professional or equivalent level*
- *Critical self-evaluation of the process*

In this section you should explain how your project deliverables and development method will meet these criteria. Some of these might be covered in other sections of the document, so refer to them if necessary.

UI/UX Mockup

At this point in the project you probably won't have a full design for whatever you are developing, but you will have a rough idea of what the UI (user interface) and UX (user experience) will be. This section should answer the key question of what your project will look like. Include a mockup of the user interface, either as a wireframe drawing or a pencil sketch that you scanned in. It doesn't have to be perfect, complete, or show every possible user interaction, and you might change the design later. It just needs to be enough to show that you have thought about what the software will look like and how it will be used.

Project plan

Include a Gantt chart (or similar project plan) with enough detail to show each stage of your project and your estimates for their duration. Show aspects that can overlap, activities that have dependencies, and so on. Remember to include time for presenting the project and writing the dissertation.

Risks & contingency plans

This section should be presented as a table with four columns: Risks, Contingencies, Likelihood and Impact. Common risks for all projects include hardware failure, software failure, running out of time, and programming problems. List possible risks in the first column, and your plans for preventing or handling them in the second column. The third column should state how likely you think the risk is (low, medium or high) and the final column should state the impact. Sometimes things with very low likelihood, such as losing your backups, could have a very high impact on the project.

References

List references to articles, websites, videos, datasets, journals, and any other background reading you carried out while writing the document and planning your project. References should be cited properly in the main body of the document. You should use the **Harvard (https://en.wikipedia.org/wiki/Parenthetical_referencing)** or **IEEE (https://en.wikipedia.org/wiki/IEEE_style)** referencing style. Guidance is available on the **university library website (https://libguides.liverpool.ac.uk/referencing)**.